NHL Regulation Time Game-Level Prediction Modelling

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1 Introduction

In this paper, we construct a prediction model to determine outcomes for NHL matches in regulation time. The prediction model works at three levels based on the three possible outcomes: 'Home Win', 'Draw' and 'Away Win'. The forecasting models implemented and tested in this paper are based on detailed NHL player-game and match-level data over multiple seasons (2011-2023). The motivation for this research lies in the value of reliable forecasts in performance analysis, pre-match preparation, team strength evaluations and for assessing whether model-based probabilities can outperform betting-market implied forecasts. Model evaluation prioritises Brier score and Log Loss, which reward good probabilities and penalise bad probabilities, to determine the strength of each model. Subsequently, the strongest model is used to construct a simple positive-expected-value betting strategy and evaluate its profitability against bookmaker odds.

The structure of the paper follows the modelling–evaluation pipeline of Holmes and McHale (2024), whose emphasis on out-of-sample probability scoring and betting-strategy performance informed both the modelling framework and the evaluation design. The also paper tests a Dixon and Coles (1997) inspired Poisson model with exponential weighting to mirror time decay and the τ-correction for low-scoring games ((0.0), (0,1), (1,0), (1,1)). In parallel, we construct an Hvattum and Arnzten (2010) inspired ordered-logit model using ELO differences and extend it by implementing a grid-search tuning of $K0, \lambda$ and the home advantage H. Finally, Peeters (2018) serves as a foundation in this paper's approach to evaluating betting strategies using model-derived probabilities and the inclusion of the positive expected value rule.

2 Data

For this assignment, two datasets were provided. The first dataset, which we refer to as the player dataset, contains information on each player–team–game combination. Each row describes the performance of a specific player in a specific match, providing a granular view of player-level contributions throughout the season. Although the dataset contains a large number of variables, only a subset is directly relevant for our analysis. The key variables used in our empirical work are:

- **teamname**: The team for which the player appeared, recorded using standard NHL abbreviations (e.g., BOS).

- **home**: A binary indicator equal to 1 if the player's team was the home team and 0 otherwise.
- **position**: The player's on-ice position (C, L, R, D, or G).
- **goals**: The number of goals scored by the player in the given match.

The second dataset, referred to as the game dataset, contains information at the match level rather than the player level. Each row corresponds to a single game played between two teams. The dataset includes standard match descriptors such as home-team indicators, team abbreviations, and goals scored in regulation time. Importantly, the dataset also reports model-implied probabilities for each of the three possible outcomes after regulation: home win, draw, and away win. These probabilities are later used in the evaluation of model performance and betting strategies.
To build intuition about the underlying data, we begin with a brief exploratory analysis. Figures 1, 2, and 3 display several descriptive plots that summarize outcome frequencies and goal distributions.

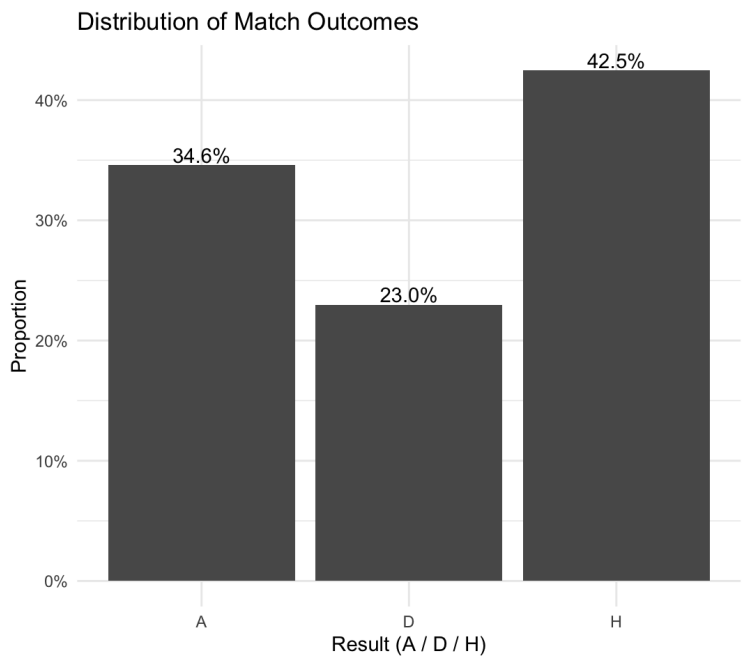

Figure 1: Overall probabilities of a match ending in an away win (A), draw (D), or home win (H).

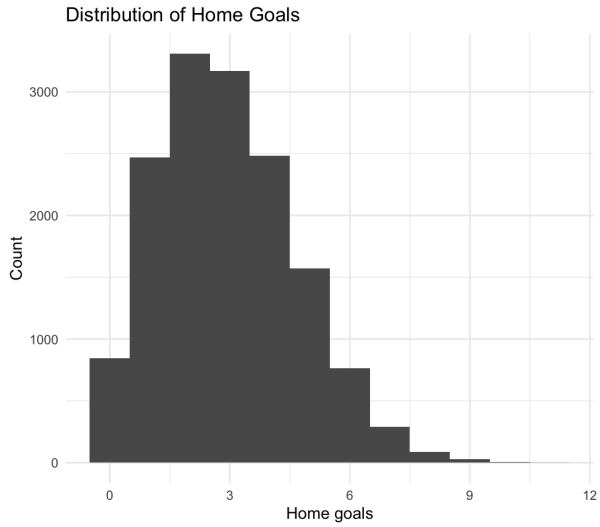

Figure 2: Distribution of goals scored by home teams.

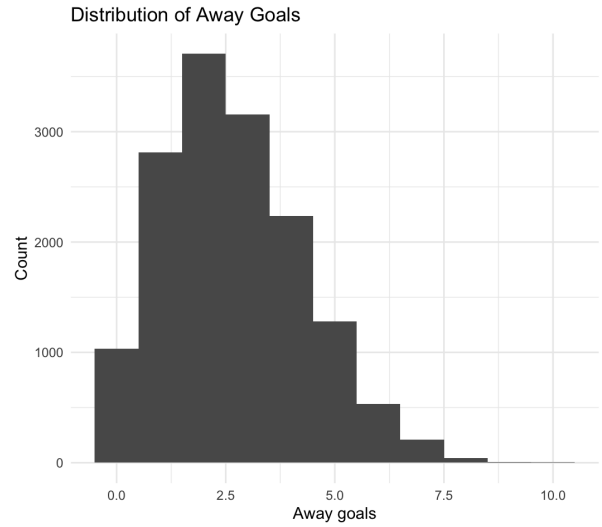

Figure 3: Distribution of goals scored by away teams.

The distributions in Figures 2 and 3 indicate that goal scoring is moderately dispersed and exhibits slight right skewness, consistent with typical scoring patterns in professional hockey. Table 1 summarizes several key descriptive statistics. The average home team scores approximately 2.93 goals per game, whereas the average away team scores around 2.67. Variances are comparable across home and away teams, suggesting similar levels of scoring volatility. The maximum number of goals scored in a single game is 11 for home teams and 10 for away teams, highlighting the presence of relatively high-scoring outliers.

Statistic	**Mean Home Goals**	**Mean Away Goals**	**Var Home Goals**	**Var Away Goals**
Value	2.929965	2.674389	2.964203	2.647699

Table 1: Summary statistics of goals scored by home and away teams.

3 Methodology

3.1 Feature Construction and Data Transformations

The dataset 'playergame_raw' is aggregated at a team level to construct the 'teamdata_gamelvl' dataset. The raw dataset represented a player-team-game combination per row, but each row in the aggregated dataset represents per-game data at a team level. Important variables in this dataset are goals, shots, plusminus, time on ice (TOI in seconds) and a newly constructed variable of TOI by position. Furthermore, game-level outcome variables, such as result, home/away goals, and goal difference, were added to the dataset to make it model-ready.

Rolling variables were constructed to capture short-term pre-match form. The optimal window

was selected through a grid search where the candidate with the lowest log loss for the relevant model was chosen. Mean shots taken by the team, goal difference with respect to the team, and mean positional time on ice were some rolling variables constructed to capture form, momentum and assess short-term team strength.

Another dataset was constructed to track the NHL game schedule. Variables to track days since last home/away game and the number of home and away games in the last 7 days were constructed to account for the effect of fatigue on match outcome, and were partially inspired by schedule considerations in Holmes and McHale (2024). In this paper, the difference variables were used as predictors to reduce multicollinearity concerns, to account for the relevance of relative strength and to make the models invariant to coding of home/away.

3.2 Rolling-window tuning and candidate selection

The rolling window was tuned using a grid search where the following candidates, $k \in (3, 5, 7, 10, 15)$, were tested. The candidates were trained on data from $2011 - 2018$ and validated on $2019 - 2020$ data. A multiclass log-loss function was used to determine the strength of each window on the overall predictions. Difference variables were enacted again for the previously stated reasons. To select the optimal k, an ordered logit model was fit on each k, log-loss was computed on the validation set, and the k that minimises the measure was chosen.

3.3 ELO-rating construction and Hyperparameter Tuning

To capture the dynamics of team strength, an ELO rating system was implemented following the structure of Hvattum and Arntzen (2010). Each team has a baseline rating of 1500 at the start of the season, which are updated sequentially after every match. The update rule depends on the expected result implied by current ratings, the actual match outcome and a goal-difference-adjusted learning rate.

For a match between home team i and away team j, the expected home win probability is computed using the standard logistic transformation, with an additional home advantage parameter H. Let R_i and R_j denote pre-match ratings for the home and away teams, respectively. Then the expected home score is:

$$E_i = \frac{1}{1 + 10^{-\frac{(R_i + H - R_j)}{400}}}.$$

The rating upgrade features a goal-difference scaling factor, allowing the magnitude of the update to increase in line with the margin of victory. If $\Delta = |G_i - G_j|$ is the regulation goal difference and $S_i \in (0, 0.5, 1)$ is the actual match score for the home team $(loss, draw, win)$, the effective learning

rate is defined as:

$$K_{\text{adj}} = K_0(1+\Delta)^{\lambda},$$

and the ELO update for the home team becomes:

$$R_i^{\text{new}} = R_i^{\text{old}} + K_{\text{adj}}(S_i - E_i),$$

with an analogous update for the away team.

Since ELO performance depends decisively on the choice of tuning parameters, a three-dimensional grid search was conducted over the base learning rate K_0, the goal-difference exponent λ, and the home-advantage parameter H. For each candidate parameter tuple (K_0, λ, H), the rating system was recomputed chronologically for the full dataset, and an ordered logit of the form

$$\Pr(Y = k \mid X) = \text{logit}^{-1}(\alpha_k - \beta \cdot \text{elo_diff}), \qquad k \in \{\text{A}, \text{D}, \text{H}\}.$$

was fitted on a trainng window (2011-2018) and evaluated on a valuation window (2019-2020). Model performance was measured by a multiclass Brier score, following recommendations from Holmes and McHale (2024). The optimal tuning paramters were selected as those that minimised the out-of-sample multiclass Brier score. This procedure yielded the paramets $K_0 = 5, \lambda = 0.5$,and $H = 30$

3.4 Model Specifications

3.4.1 Ordered Logit - Rolling + TOI

This model uses short-term team performance to explain regulation time outcomes. Rolling average of shots Rolling averages of shots, goal difference, win percentage, and positional time-on-ice are computed over the optimal window length k, and converted to home-away differences prior to estimation.

$$\begin{aligned}\Pr(Y = k \mid X) = \text{logit}^{-1}\Big(&\alpha_k - \beta_1\,\text{shots_diff}_t - \beta_2\,\text{gd_diff}_t - \beta_3\,\text{win_diff}_t \\ &-\beta_4\,\text{toi_fwd_diff}_t - \beta_5\,\text{toi_def_diff}_t\Big), \qquad k \in \{\text{A}, \text{D}, \text{H}\}.\end{aligned} \tag{1}$$

This transformation helps capture relative team strength entering each match and avoids multicollinearity between home and away team features. The ordered-logit specification models the latent advantage of the home team as a linear function of these rolling differences, with the thresholds determing the probability of win for Home or Away, or that of a Draw.

3.4.2 Ordered Logit - ELO only

The ELO-only model uses the pre-match ELO differential as the sole predictor of match outcomes. ELO ratings summarise long-term team strength and are updated sequentially after each game. The

ordered-logit framework interprets higher ELO difference as the increasing the latent probability of a home win. This serves as suitable comparison for comparing dynamic rating-based models to richer feature-based models.

$$\Pr(Y = k \mid X) = \text{logit}^{-1}\big(\alpha_k - \gamma\,\text{elo_diff}_t\big), \qquad k \in \{\text{A}, \text{D}, \text{H}\}. \tag{2}$$

3.4.3 Ordered Logit - ELO + Rest

This specification adapts the ELO difference with rest-day and schedule-congestion differences, to capture the effects of fatigue as documented in relevant literature. The variable $rest_diff_t$ and $games_last7_diff_t$ measure whether one team enters the match more rested than its opponent. Hence, this model accounts for both long-term strength and short-term physical disadvantage.

$$\begin{aligned}\Pr(Y = k \mid X) = \text{logit}^{-1}\Big(\alpha_k - \gamma\,\text{elo_diff}_t - \delta_1\,\text{rest_diff}_t - \delta_2\,\text{games_last7_diff}_t\Big),\\ k \in \{\text{A}, \text{D}, \text{H}\}.\end{aligned} \tag{3}$$

3.4.4 Ordered Logit - ELO + Rest + Rolling + TOI

The combined model integrates long-term strength (ELO), short-run performance (rolling form), and player usage/scheduling information (TOI and rest). It is the most comprehensive specification as rolling differences capture recent momentum, Elo summarises baseline ability, rest and congestion address fatigue, and TOI differences capture tactical deployment of forwards and defencemen. This was found to be the strongest overall performer in out-of-sample accuracy, Brier score, and log-loss.

$$\begin{aligned}\Pr(Y = k \mid X) = \text{logit}^{-1}\Big(&\alpha_k - \gamma\,\text{elo_diff}_t - \delta_1\,\text{rest_diff}_t - \delta_2\,\text{games_last7_diff}_t - \beta_1\,\text{shots_diff}_t\\ &-\beta_2\,\text{gd_diff}_t - \beta_3\,\text{win_diff}_t - \beta_4\,\text{toi_fwd_diff}_t - \beta_5\,\text{toi_def_diff}_t\Big), \qquad k \in \{\text{A}, \text{D}, \text{H}\}.\end{aligned} \tag{4}$$

3.4.5 Poisson Goal Models (Time-Decayed Dixon Coles

The Poisson model follows the Dixon and Coles (1997) framework, modelling home and away goals as separate Poisson processes. Rolling attacking and defensive statistics enter each log-mean equation, along with the Elo differential. To account for the declining relevance of older matches, exponential time-decay weights are applied during estimation. Additionally, the Dixon–Coles τ -correction is applied to adjust joint goal probabilities for low-scoring dependence. The resulting bivariate goal distribution is summed over scorelines to obtain probabilities of Away win, Draw, and Home win.

Home Goals Model

$$G_{i,t}^{\text{home}} \sim \text{Poisson}(\lambda_{i,t}^{H}),$$

$$G_t^{H} \sim \text{Poisson}(\lambda_{H,t}),$$

$$\log(\lambda_{H,t}) = \alpha_H + \beta_1 \,\text{shots_lastk_home} + \beta_2 \,\text{goals_lastk_home} + \beta_3 \,\text{goals_conceded_lastk_away} \\ + \beta_4 \,\text{shots_conceded_lastk_away} + \beta_5 \,\text{elo_diff}_t. \tag{5}$$

Away Goals Model

$$G_t^A \sim \text{Poisson}(\lambda_{A,t}),$$

$$\log(\lambda_{A,t}) = \alpha_A + \beta_1 \,\text{shots_lastk_away} + \beta_2 \,\text{goals_lastk_away} + \beta_3 \,\text{goals_conceded_lastk_home} \\ + \beta_4 \,\text{shots_conceded_lastk_home} + \beta_5 \,\text{elo_diff}_t. \tag{6}$$

Exponential time-decay weight

$$w_t = \exp(-\xi \, d_t),$$

Dixon-Coles τ correction

$$\tau(gh, ga; \rho) = \begin{cases} 1 - \rho, & (gh, ga) \in \{(0,0), (1,1)\}, \\ 1 + \rho, & (gh, ga) \in \{(0,1), (1,0)\}, \\ 1, & \text{otherwise.} \end{cases}$$

Poisson score to outcome probabilites

$$P(\text{Home win}) = \sum_{gh > ga} \tau(gh, ga; \rho) \; \text{Pois}(gh \mid \lambda_H) \; \text{Pois}(ga \mid \lambda_A),$$

$$P(\text{Draw}) = \sum_{gh = ga} \tau(gh, gh; \rho) \; \text{Pois}(gh \mid \lambda_H) \; \text{Pois}(gh \mid \lambda_A),$$

$$P(\text{Away win}) = \sum_{gh < ga} \tau(gh, ga; \rho) \; \text{Pois}(gh \mid \lambda_H) \; \text{Pois}(ga \mid \lambda_A).$$

3.4.6 Bookmaker Comparison and Betting Strategy

To evaluate whether statistical forecasts outperform the market, bookmaker decimal odds for Home win, Draw and Away win are incorporated using the market data from *gamedata.csv*. For each match, the odds $odds_k$ implies a set of bookmaker-implied probabilities:

$$p_k^* = \frac{1/\text{odds}_k}{\sum_{j \in \{\text{H,D,A}\}} 1/\text{odds}_j}, \qquad k \in \{\text{H, D, A}\}.$$

which account for the bookies margin.

Using the probabilites p_k generated by the statistical model, the expected value (EV) of betting on outcome k is:

$$\mathrm{EV}_k = p_k \,\mathrm{odds}_k - 1, \qquad k \in \{\mathrm{H}, \mathrm{D}, \mathrm{A}\},$$

For each match, the strategy selects the outcome with the highest EV and places a bet only when this EV is strictly positive. This is similar to exploiting only those situations in which the model-implied probability exceeds the implied market probability. A fixed stake of one unit is used for all positive-EV bets.

Performance of the betting strategy is evaluated over the $2021 - 2023$ test window using total profit, return on investment (ROI), and the Sharpe ratio, following Holmes and McHale (2024).This method grants a meaningful way to evaluate the ability of forecasting models to identify exploitable market inefficiencies.

4 Model Performance

4.1 Evaluation Framework

Model performance is evaluated using the multiclass Brier score, multiclass accuracy, and Log Loss. Following the evaluation strategy of Holmes and McHale (2014),the emphasis is placed on how accurately predicted probabilities reflect real world outcomes. All estimation was carried out on 2011-2020 data and evaluated strictly out of sample based on matches from 2021 to 2023. Rolling variables, rest indicators, and ELO ratings were updated sequentially to ensure no future information entered the prediction window, so only information available prior to each match was used.

4.2 Performance by model

Table 2: Predictive performance of binary and multiclass models on the 2021–2023 test set.

Model	Log Loss	Brier score	Accuracy
Multiclass classification (W/D/L)			
Naive benchmark (Always H)	1.067	0.647	0.485
Ordered Logit (ELO only)	1.030	0.619	0.493
Ordered Logit (TOI + Rolling)	1.038	0.625	0.485
Ordered Logit (ELO + Rest)	1.030	0.619	0.492
Ordered Logit (Combined)	1.027	0.617	0.499
Poisson (Goals $\rightarrow$ W/D/L)	1.04	0.623	0.495

The predictive results for $2021-2023$ test period are summarised in Table 2. Across all models, the multiclass ordered-logit specifications outperform the naive 'always home win' benchmark in both log loss and Brier score, providing evidence of better calibrated probability forecasts relative to the

trivial baseline. Models using only Elo or only rolling/TOI features achieve similar performance, while the inclusion of rest-related variables provides modest but consistent improvements.

The strongest overall specification is the Ordered Logit (combined) model. which integrates Elo ratings, rest-day differences, rolling form, and TOI usage. It attains the lowest log-loss (1.027) and lowest Brier score (0.617), as well as the highest classification accuracy (0.499). Although the gains over simpler models are minimal, the combined model demonstrates the most reliable out-of-sample performance and is therefore selected as the preferred forecasting model for subsequent betting-strategy analysis.

4.3 Coefficient estimation

For sake of simplicity, we only report the estimated coefficients of the model we deemed best, which is the ordered logit (ELO + Rest + Rolling). It perfomed the best among our tested models, but more on this in 5. The rest of our estimations for other models can be found in 7.

Variable	**Estimate**	**Std. Error**	**t-value**
elo_diff	0.0029969	0.0003251	9.2192
rest_diff	0.0024603	0.0014541	1.6920
games_last7_diff	-0.0374425	0.0211789	-1.7679
shots_diff_lastk	0.0225608	0.0057309	3.9367
gd_diff_lastk	0.0795512	0.0458684	1.7343
win_diff_lastk	0.0381664	0.2079429	0.1835
toi_fwd_diff_lastk	0.0001002	0.0001440	0.6955
toi_def_diff_lastk	0.0001699	0.0001577	1.0772
goals_conceded_diff_lastk	0.0516124	0.0429471	1.2018
shots_conceded_diff_lastk	-0.0367987	0.0054994	-6.6914

Table 3: Coefficient Estimates from Ordered Logit Model (ELO /+ Rol

4.4 Betting Strategy Performance

Metric	**Value**
Number of bets placed	3763
Total profit	29.36
Return on Investment (ROI)	0.007802285
Sharpe ratio	0.004457025

Table 4: Performance of the EV-based fixed-stake betting strategy.

Table 4 reports the performance of the positive-expected-value betting strategy applied to the $2021 - 2023$ test period. The strategy placed 3763 bets, each with a fixed stake of one unit, whenever the model-implied expected value of an outcome exceeded zero. Across all bets, the cumulative

profit was 29.36 units, corresponding to a return on investment (ROI) of approximately 0.78%.

Although the strategy produced a small overall profit, the associated Sharpe ratio (0.0045) indicates that returns were extremely volatile relative to their mean. That is, the economic profit is economically negligible once risk is accounted for, and the low Sharpe ratio suggests that any apparent outperformance of the market is not statistically meaningful. Overall, while the model's forecasts yields positve expected value, the results do not provide strong evidence of exploitable or persistent inefficiencies in NHL betting markets.

5 Conclusion

This report developed and evaluated several models designed to predict NHL match outcomes at the end of regulation time. To prevent overfitting or information leakage, a strict chronological split was used to ensure the model selection was based entirely on information available prior to the evaluation period.

Among all specifications, the Ordered Logit model combining ELO ratings, rest variables, and rolling performance indicators delivered the strongest overall results. It achieved the lowest multiclass Brier score (0.617) and the highest accuracy (0.499).This indicates that it produces both sharp and well-calibrated probabilities.

The performance patterns across models highlight two clear insights: ELO ratings provide the most stable measure of underlying team strength, and they become even more effective when supplemented with short term form and rest related features. Models relying only on rolling variables were noticeably weaker, while the Poisson model offered reasonable accuracy, but inferior calibration.

Despite these results, some limitations remain. Information on player level, lineup uncertainty, and more detailed team metrics were outside the scope of this assignment, but could enhance the model´s accuracy in future research. Another big limitation is the relatively small size of the test set, as it only covers three seasons. A larger sample size consisting of more seasons and additional leagues would provide a more reliable indication of the robustness of the model.

Overall, the Ordered Logit (ELO + Rest + Rolling) specification provides a robust and reliable foundation for prediction and will be used for the betting strategy that follows. It´s stability in the out-of-sample period makes it the most reliable choice among the models tested.

6 AI Reflection

AI was primarily useful to structure the skeleton of our approach. The models were decided on based on our own understanding and research, however, AI was useful in directing us to relevant libraries to ease the implementation of the models in R. Considering numerous different models were tested, AI was useful in navigating the dplyr framework to create new variables or alter the dataset without any errors. It was also used to help write the README file, to ensure that the provided replication steps are thorough.

During the construction of the Dixon-Coles Poisson models, AI recommended the use of the stan library to get the most useful implementation. However, the understanding of the suggestions was beyond our scope and was rejected. Hence, our final Poisson implementation borrows elements from Dixon and Coles (1997), but is not an exact remake. Finally, AI helped format the LaTeX document to write the necessary already-implemented formulas and align any figures and tables.

7 Appendix

Variable	Estimate	Std. Error	t-value
shots_diff_lastk	3.220e-02	0.0056218	5.7282
gd_diff_lastk	1.500e-01	0.0451513	3.3211
win_diff_lastk	3.123e-01	0.2053271	1.5209
toi_fwd_diff_lastk	8.992e-05	0.0001438	0.6254
toi_def_diff_lastk	2.189e-04	0.0001574	1.3903
goals_conceded_diff_lastk	4.978e-02	0.0428349	1.1621
shots_conceded_diff_lastk	-4.438e-02	0.0054334	-8.1674

Table 5: Coefficient Estimates for Ordered Logit (Rolling + TOI)

Coefficient	Value	Std. Error	t-value
elo_diff	0.004355	0.0002437	17.87

Table 6: Regression Coefficient for ordered logit using ELO only

Coefficient	Estimate	Std. Error	z-value	Pr(¿—z—)
(Intercept)	0.0345561	0.1383111	0.250	0.8027
goals_lastk_home	0.0341806	0.0175830	1.944	0.0519
shots_lastk_home	0.0157712	0.0034159	4.617	3.89e-06
goals_conceded_lastk_away	0.0069675	0.0181072	0.385	0.7004
shots_conceded_lastk_away	0.0140045	0.0033560	4.173	3.01e-05
elo_diff	0.0006792	0.0001168	5.815	6.06e-09

Table 7: Poisson

Coefficient	Value	Std. Error	t-value
elo_diff	0.004376	0.0002439	17.942
rest_diff	0.002370	0.0014470	1.638
games_last7_diff	-0.035793	0.0211310	-1.694

Table 8: ELO + Rest